Table 17.2 Auto Scaling group parameters

Name	Description	Possible values
`DesiredCapacity`	Desired number of healthy virtual machines	Integer
`MaxSize`	Maximum number of virtual machines; the upper scaling limit	Integer
`MinSize`	Minimum number of virtual machines; the lower scaling limit	Integer
`HealthCheckType`	How the Auto Scaling group checks the health of virtual machines	`EC2` (health of the instance) or `ELB` (health check of instance performed by a load balancer)
`HealthCheckGracePeriod`	Period for which the health check is paused after the launch of a new instance to wait until the instance is fully bootstrapped	Number of seconds
`LaunchTemplate`	ID (`LaunchTemplateId`) and version of launch template used as a blueprint when spinning up virtual machines	ID and version of launch template
`TargetGroupARNs`	The target groups of a load balancer, where autoscaling registers new instances automatically	List of target group ARNs
`VPCZoneIdentifier`	List of subnets in which to launch EC2 instances	List of subnet identifiers of a VPC

If you specify multiple subnets with the help of `VPCZoneIdentifier` for the Auto Scaling group, EC2 instances will be evenly distributed among these subnets and, thus, among availability zones.

Don’t forget to define a health check grace period

If you are using the ELB’s health check for your Auto Scaling group, make sure you specify a `HealthCheckGracePeriod` as well. Specify a health check grace period based on the time it takes from launching an EC2 instance until your application is running and passes the ELB’s health check. For a simple web application, a health check period of five minutes is suitable.

The next listing shows how to set up such a dynamic EC2 instance pool with the help of a CloudFormation template.